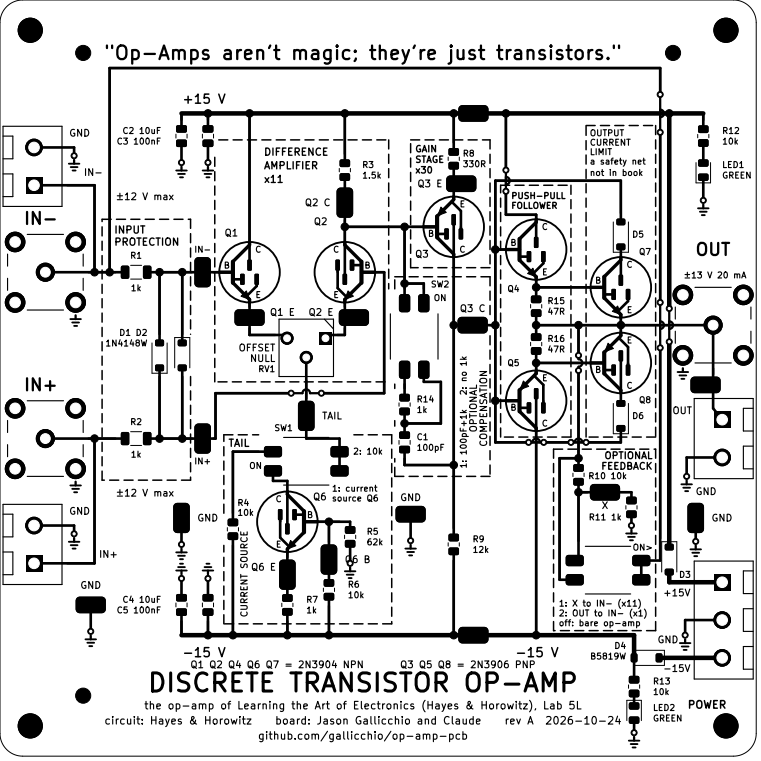

"Op-Amps aren't magic; they're just transistors."
+15 V
GND
C2 10uF
C3 100nF
IN-
±12 V max
INPUT PROTECTION
R1
1k
D1 D2
1N4148W
R2
1k
IN+
DIFFERENCE AMPLIFIER x11
Q1
Q2
Q1 E
Q2 E
OFFSET NULL RV1
R3
1.5k
Q2 C
GAIN STAGE x30
Q3 E
Q3
Q3 C
R8
330R
SW2
ON
R14
1k
C1
100pF
1: 100pF 1k
2: no 1k
OPTIONAL COMPENSATION
TAIL
SW1
2: 10k
1: current source Q6
Q6
R4
10k
R5
62k
Q6 B
R6
10k
Q6 E
R7
1k
CURRENT SOURCE
R9
12k
PUSH-PULL FOLLOWER
Q4
Q5
R15
47R
R16
47R
OUTPUT CURRENT LIMIT
a safety net not in book
D5
Q7
Q8
D6
OPTIONAL FEEDBACK
R10 10k
R11 1k
1: X to IN- (x11)
2: OUT to IN- (x1)
off: bare op-amp
R12
10k
LED1
GREEN
OUT
±13 V 20 mA
D3
+15V
D4
B5819W
R13
10k
LED2
GREEN
POWER
-15V
-15 V
Q1 Q2 Q4 Q6 Q7 = 2N3904 NPN
Q3 Q5 Q8 = 2N3906 PNP
C4 10uF
C5 100nF
DISCRETE TRANSISTOR OP-AMP
the op-amp of Learning the Art of Electronics (Hayes & Horowitz), Lab 5L
circuit: Hayes & Horowitz board: Jason Gallicchio and Claude rev A 2026-10-24
github.com/gallicchio/op-amp-pcb